

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF SOFTWARE ENGINEERING

FIRST YEAR - BATCH 2024

MID TERM EXAMINATIONS – SPRING 2024

Duration: 90 Minutes

Date: 22-05-2024

Max. Marks: 20

ES-205 (Islamic Studies)

Note: Attempt all of the given questions. Read the questions carefully and answer accordingly.

Q1. Describe essential details about the Event of Ascension. Also, share your understanding of the event of Ascension. And make a biographical chart from 4th year of prophethood to 13th year of prophethood. 9

CLO# 2 (10 Marks)

Q2. Translate all of the sacred texts and discuss the concept of Tauheed and different aspects in the light of the given texts of the holy Quran. 9.5

CLO# 1 (10 Marks)

1. لَوْ كَانَ فِيهِمَا آلِهَةٌ إِلَّا اللَّهُ لَفَسَدَتَا فَسُبْحَانَ اللَّهِ رَبِّ الْعَرْشِ عَمَّا يَصِفُونَ ﴿٢٢﴾ (الأنبياء: 22)
2. وَإِلَهُكُمْ إِلَهٌ وَاحِدٌ لَا إِلَهَ إِلَّا هُوَ الرَّحْمَنُ الرَّحِيمُ ﴿١٦٣﴾ (البقرة: 163)
3. إِنَّ فِي خَلْقِ السَّمَاوَاتِ وَالْأَرْضِ وَاخْتِلَافِ اللَّيْلِ وَالنَّهَارِ وَالْفُلْكِ الَّتِي تَجْرِي فِي الْبَحْرِ بِمَا يَنْفَعُ النَّاسَ وَمَا أَنْزَلَ اللَّهُ مِنَ السَّمَاءِ مِنْ مَاءٍ فَأَخْيَا بِهِ الْأَرْضَ بَعْدَ مَوْتِهَا وَبَثَّ فِيهَا مِنْ كُلِّ دَابَّةٍ وَتَصْرِيفِ الرِّيَّاحِ وَالسَّحَابِ الْمُسَخَّرِ بَيْنَ السَّمَاءِ وَالْأَرْضِ لَآيَاتٍ لِقَوْمٍ يَعْقِلُونَ ﴿١٦٤﴾ (البقرة: 164)

18.5
20

TIME: 1.5 hours

DATED: 20/05/2024

MAX MARKS: 20

ACADEMIC READING AND WRITING (EA-115)**Note: Attempt all questions in your answer sheets.****Q.1. Read the passage below and answer the questions under it. [CLO 1, 8 marks]****THE KOREAN WAVE**

The Asia Society Korea Center's Monthly Luncheon returned on Tuesday with a fascinating insight into Korean pop music from the President of DFSB Kollektive, Bernie Cho. Mr. Cho specializes in providing digital media, marketing, and distribution solutions to over 350 Korean Pop music artists and was kind enough to speak at the Korea Center on "A Case Study on K-Pop: The Hip Hype Reality of the Korean Wave". Mr. Cho first came to Korea in 1993 with the goal of pursuing his graduate studies at Yonsei University, however through a chance meeting at a Seoul nightclub ended up working on the launch of Korea's first music channel and his career in the music industry began. His talk focused on how the successes of the Korean music industry have stemmed from the frustrations those within the business have faced and how through diversification and glocalization they were able to flip these challenges and turn them into opportunities.

Mr. Cho explained how in 1993 the Korean music industry was very different from the one we see today; 70% of music in the country was international pop music with Korean trot and country music making up the other 30%. Fast forward to 2002 and K-Pop, as it was now known, accounted for 70% of the market, and Korean artist BoA was dominating the charts both in Korea and Japan. Today, Korea is ranked as the 8th largest music market in the world, was the first country in the world where digital sales outsold physical records, and is the 4th largest net music exporter behind the U.S., U.K., and Sweden. Mr. Cho talked about how these accolades came as a result of hard work and sweat born out of frustrations that the Korean music industry constantly faced.

Korea is the most wired nation in the world where cell phone penetration stands at 110% and the country is experiencing the digital age. Yet the digital music stores and services are dominated by Korean providers where overseas services such as iTunes and Spotify are prohibited. Mr. Cho explained how the problem lies in the price that these Korean portals charge for downloads; at 9 cents a song, which is 16 times cheaper than in the U.S and 32 times cheaper than in Japan, a lob-sided market has formed where a song made in Korea can make 23% more profit when sold overseas. Because of these market conditions, the Korean music industry has had to diversify and look at other opportunities to make money.

One opportunity lies in the overseas market where Korea has seen over 60% growth in exports over the past 5 years. Mr. Cho gave the example of Psy and his worldwide hit Gangnam Style as a benefit of tapping into the overseas marketplace. In Korea, the song had 3.6 million digital downloads and over 40 million streams, yet earned Psy only \$61,000 compared to the \$ 2.7 million he earned from roughly the same number of downloads and streams outside of Korea. S.M Entertainment and YG Entertainment, two of Korea's biggest music companies only produce 20-30% of revenue from music sales in Korea; more than 50% comes from overseas.

So why has Korea been so successful in the overseas market? Mr. Cho puts this down to "glocalization". The early 2000s saw a shift from Korean-made music for domestic consumption to Korean-made music marketed for international consumption. Today we even see music made by Koreans but written, recorded, and performed in countries such as the U.S and Sweden. Three factors have helped with this "glocalization"; the bilingual music being produced, multicultural K-pop groups with international members, and a boom in K-pop fan sites that dominate online portals.

Through the domestic challenges that those in the industry faced, Korean entrepreneurs were forced to explore other avenues to generate profit, of which the overseas market played a huge role. Mr. Cho summarized his talk by saying that without the influence of these Korean music entrepreneurs, the "Korean Wave" could well have turned into a "Korean Tsunami" and we would not have the booming industry we have today.

1. What was the state of the Korean music industry in 1993 and how has it changed over the years?

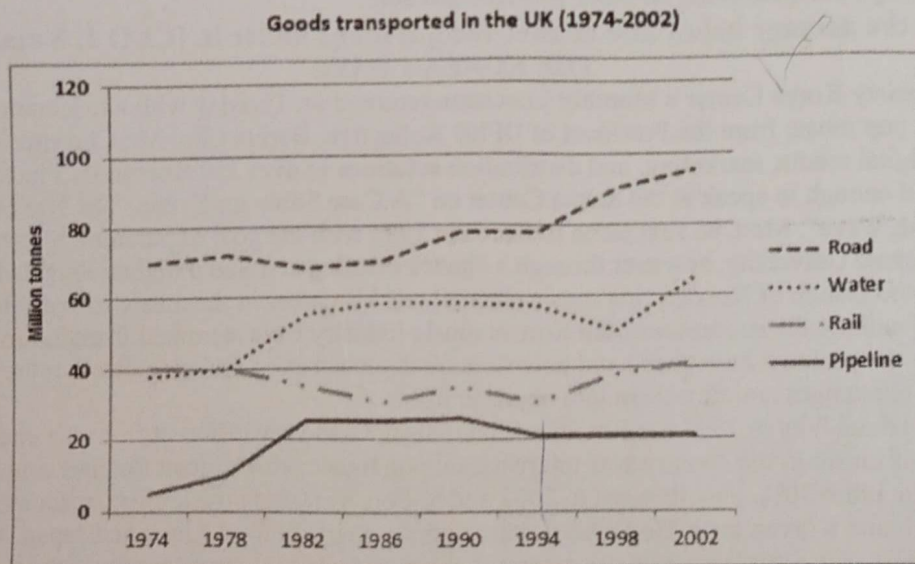
NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FE SEC A & B (BACHELOR OF SOFTWARE ENGINEERING)

SPRING SEMESTER MIDTERM EXAMINATIONS 2024

2. What are the three factors that have helped with the "glocalization" of Korean music?
3. Which Korean music artists have contributed to the Korean economy and how?
4. Describe another movement like the Korean Wave in any region that has impacted the social, cultural, political, or economic condition of the region.

Q.3. Discuss the weight of the goods transported in the UK across the years stated along the x-axis, in the visual below. [CLO 1, 4 marks]



Q.3. Brainstorm ideas on any 1 of the topics provided below using any one *relevant* pre-writing strategy. [CLO 2, 2 marks]

- Reading fiction books can enhance our view of the world. Do you agree or disagree?
- Lower grades should be punished. Do you agree or disagree?
- Homework tasks help students retain knowledge better. Do you agree or disagree?

Q.4. Write a 200-word (approx.) essay using the ideas you have brainstormed in Q.3. [CLO 2, 4 marks]

Q.5. Highlight the thesis and forecasting statements and the topic sentences of each of the body paragraphs. [CLO 2, 2 marks]

Best of Luck!

Software Engineering - SE-207

Instructions: Attempt All Questions

Q#1.

CLO4

a. You are leading a team tasked with developing a new mobile application for a startup company. Given the dynamic nature of the project requirements and the need for early user feedback, which software process model would you recommend: the Spiral model or the Prototype model? **Analyze** the project requirements and **discuss** how each model addresses the project's challenges and constraints. Provide examples of specific scenarios where each model would excel in ensuring successful project outcomes. [06]

b. Some people argue that developers should not be involved in testing their own code but that all testing should be the responsibility of a separate team. **Give** arguments for and against testing by the developers themselves. [05]

Q#2.

CLO2

Design a reservation system for a movie theater. **Draw** a use case diagram and a sequence diagram for the process of making a reservation through the system.

a. Use Case Diagram: **Illustrate** the main actors involved in the system, identify key use cases related to the reservation process, including "Browse Movies," "Select Showtime," "Reserve Seats," "Make Payment," and "Cancel Reservation". Show relationships between actors and use cases, indicating which use cases each actor can initiate or participate in. [06]

b. Sequence Diagram: **Draw** a sequence diagram specifically for the "Make Reservation" process. Identify the actors involved, sequence the interactions between these actors and the system components during the reservation process and include messages exchanged, sequence of actions and any conditional logic or decision. [06]

Q#3.

CLO1

a. You have been given the responsibility to elicit requirements from a customer who informs you he is too busy to meet. **State** and explain how you would deal with such situation. [03]

b. **Explain** how a program can be correct and still not be reliable. [03]

c. **State** why a highly coupled module difficult to unit test? [03]

d. **List** and give one line description of Gravin's 8 quality dimensions. [03]

Q#4.

CLO1

Describe the concept of the 4 Ps in software project management. Discuss how each 'P' contributes to the success of a software project development. Provide examples or scenarios where the effective management of each 'P' would be crucial for achieving project objectives and ensuring stakeholder satisfaction. [08]

Q#5.

CLO4

Defend incremental testing strategy for testing an e-commerce website and **discuss** how it can be implemented to test the site. Also, **compare** black box and white box testing techniques. [09]

Q#6.

CLO2

a. Considering each of the four aspects of the cost of quality, **choose** the most expensive one and explain why? [04]

b. Do you agree or disagree with the following statement: "Since we deliver multiple increments to the customer, why should we be concerned about quality in the early increments—we can fix problems in later iterations." **Explain** and justify your answer. [04]

Seat No. 54 - 23074

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
FIRST YEAR (PETROLEUM/ COMPUTER SYSTEMS/ SOFTWARE/ APPLIED PHYSICS/
MANAGEMENT SCIENCE)
SPRING SEMESTER EXAMINATIONS 2024
Batch - 2023

Time: 3 Hours

Dated: 26-07-2024

Max.Marks: 60

ISLAMIC STUDIES - ES-205

Instructions: Read the question paper carefully, and answer all the questions below.

Q.1 Translate the given verses. And explain the concept of here-after in light of both texts in the holy Quran. CLO# 1 (10 Marks)

وَاتَّقُوا يَوْمًا لَا تَجْزِي نَفْسٌ عَنْ نَفْسٍ شَيْئًا وَلَا يَقْبَلَ مِنْهَا شَفَاعَةٌ وَلَا يُؤْخَذُ مِنْهَا عَدْلٌ وَلَا هُمْ يُنصَرُونَ ﴿البقرة: 48﴾

يَا أَيُّهَا النَّاسُ إِنَّ كُنْتُمْ فِي رَيْبٍ مِنَ الْبَيْتِ فَإِنَّا خَلَقْنَاكُمْ مِنْ تُرَابٍ ثُمَّ مِنْ نُطْفَةٍ ثُمَّ مِنْ عَلَقَةٍ ثُمَّ مِنْ مُضْغَةٍ مُخَلَّقَةٍ وَغَيْرِ مُخَلَّقَةٍ لِنَبَيِّنَ لَكُمْ وَنَقَرُ فِي الْأَرْحَامِ مَا نَشَاءُ إِلَى أَجَلٍ مُسَمًّى ثُمَّ نُخْرِجُكُمْ طِفْلًا ثُمَّ لِتَبْلُغُوا أَشُدَّكُمْ وَمِنْكُمْ مَنْ يُتَوَفَّى وَمِنْكُمْ مَنْ يُرَدُّ إِلَى أَرْذَلِ الْعُمُرِ لِكَيْلَا يَعْلَمَ مِنْ بَعْدِ عِلْمٍ شَيْئًا وَتَرَى الْأَرْضَ هَامِدَةً فَإِذَا أَنْزَلْنَا عَلَيْهَا الْمَاءَ اهْتَزَّتْ وَرَبَتْ وَأَنْبَتَتْ مِنْ كُلِّ زَوْجٍ بَهيجٍ ﴿الحج: 5﴾

Q2. (a) Explain the given text of the Holy Qur'an. CLO#1 (05-Marks)

إِنَّ الْمُسْلِمِينَ وَالْمُسْلِمَاتِ وَالْمُؤْمِنِينَ وَالْمُؤْمِنَاتِ وَالْقَانِتِينَ وَالْقَانِتَاتِ وَالصَّادِقِينَ وَالصَّادِقَاتِ وَالصَّابِرِينَ وَالصَّابِرَاتِ وَالْخَاشِعِينَ وَالْخَاشِعَاتِ وَالْمُتَصَدِّقِينَ وَالْمُتَصَدِّقَاتِ وَالصَّالِمِينَ وَالصَّالِمَاتِ وَالْحَافِظِينَ فُرُوجَهُمْ وَالْحَافِظَاتِ وَالذَّاكِرِينَ اللَّهَ كَثِيرًا وَالذَّاكِرَاتِ أَعَدَّ اللَّهُ لَهُمْ مَغْفِرَةً وَأَجْرًا عَظِيمًا ﴿الأحزاب: 35﴾

(b) Translate the given text of Hadith. CLO#1 (05-Marks)

الحلال عَنْ أَبِي عَبْدِ اللَّهِ النَّعْمَانِ بْنِ بَشِيرٍ رَضِيَ اللَّهُ عَنْهُمَا قَالَ: سَمِعْتُ رَسُولَ اللَّهِ صَلَّى اللَّهُ عَلَيْهِ وَسَلَّمَ يَقُولُ: (إِنَّ بَيْنَ وَإِنْ الْحَرَامَ بَيْنَ وَبَيْنَهُمَا أُمُورٌ مُشْتَبِهَاتٌ لَا يَعْلَمُهُنَّ كَثِيرٌ مِنَ النَّاسِ، فَمَنْ اتَّقَى الشُّبُهَاتِ فَقَدْ اسْتَبْرَأَ لِدِينِهِ وَعَرْضِهِ، وَمَنْ وَقَعَ فِي الشُّبُهَاتِ وَقَعَ فِي الْحَرَامِ كَالرَّاعِي يَرْعَى حَوْلَ الْحِمَى يُوشِكُ أَنْ يَقَعَ فِيهِ. أَلَا وَإِنْ لِكُلِّ مَلِكٍ حِمَى. أَلَا وَإِنْ حِمَى اللَّهِ مَحَارِمُهُ، أَلَا وَإِنْ فِي الْجَسَدِ مُضْغَةٌ إِذَا صَلَحَتْ صَلَحَ الْجَسَدُ كُلُّهُ وَإِذَا فَسَدَتْ فَسَدَ الْجَسَدُ كُلُّهُ أَلَا وَهِيَ الْقَلْبُ - رَوَاهُ الْبُخَارِيُّ وَمُسْلِمٌ

Q3. Translate the given verses of the holy Quran. Also explain the concept of worship in the light of surah Muminoon verses 1-11. CLO #1 (10--- Marks)

قَدْ أَفْلَحَ الْمُؤْمِنُونَ ﴿الَّذِينَ هُمْ فِي صَلَاتِهِمْ خَاشِعُونَ﴾ ﴿وَالَّذِينَ هُمْ عَنِ اللَّغْوِ مُعْرِضُونَ﴾ ﴿وَالَّذِينَ هُمْ لِلزَّكَاةِ فَاعِلُونَ﴾ ﴿وَالَّذِينَ هُمْ لِفُرُوجِهِمْ حَافِظُونَ﴾ ﴿إِلَّا عَلَى أَزْوَاجِهِمْ أَوْ مَا مَلَكَتْ أَيْمَانُهُمْ فَإِنَّهُمْ غَيْرُ مَلُومِينَ﴾ ﴿فَمَنْ اسْتَعَى وَرَاءَ ذَلِكَ فَأُولَئِكَ هُمُ الْعَادُونَ﴾ ﴿وَالَّذِينَ هُمْ لِأَمَانَاتِهِمْ وَعَهْدِهِمْ رَاعُونَ﴾ ﴿وَالَّذِينَ هُمْ عَلَى صَلَوَاتِهِمْ يُحَافِظُونَ﴾ ﴿أُولَئِكَ هُمُ الْوَارِثُونَ﴾ ﴿الَّذِينَ يَرِثُونَ الْفِرْدَوْسَ هُمْ فِيهَا خَالِدُونَ﴾

Q4. Discuss the significance of migration to Medina. Also, explain the social & political outcomes of the Treaty of Hudaibya. CLO #2 ((7+8)--- Marks)

Q5. Explain how the prophet (ﷺ) established peace in Medina. Also, describe the causes and outcomes of Ghazwa e Khandaq (Confederates). CLO #2 ((8+7)--- Marks)

--- May Allah (SWT) Enlighten you with wisdom---

SEAT NO. SE-23094**NED UNIVERSITY OF ENGINEERING & TECHNOLOGY**

FIRST YEAR(Software)

SPRING SEMESTER EXAMINATIONS 2024

Time : 3 Hours

Batch 2023

Dated : 19-JUL-24

Max Marks : 60

Applied Physics - PH-122

NOTE: ATTEMPT ALL QUESTIONS. Attempt all parts of question "together" each question carries equal marks

Q. No.	Questions	Marks	CLO
1			
a	<u>STATE</u> and <u>DERIVE</u> Work-Energy Theorem for variable forces. Does Work-Energy theorem also hold for constant force? <u>SHOW</u> your answer through mathematical steps.	4	1
b	<u>Carry out</u> the mathematical steps for displacement, velocity, acceleration, time period of Simple Harmonic Oscillator. Also <u>DERIVE</u> the equation for maximum kinetic energy, maximum potential energy and total energy for Simple Harmonic Oscillator.	4	1
c	State and prove Gauss's Law for N-charges. State its differential and integral form with mathematical equation.	4	1
d	State and explain law of conservation of angular momentum. Derive the relation between torque and angular momentum.	4	1
e	a) Compare electrostatic force and magnetic force? b) Differentiate spontaneous and stimulated emission?	4	1
2	<i>Use appropriate mathematical relation to find the following:</i>		
a	A water droplet 10^{-2} cm in diameter carries the negative charge such that the electric field at its surface is 60,000 volts/m. how strong a vertical field would be required to keep the drop from falling.	5	2
b	A mass of 0.5 kg connected to a light spring of force constant 20N/m oscillates on a horizontal friction less surface. a) Calculate the total energy of the system and Maximum speed of the mass if the amplitude of motion is 3cm. b) What is the velocity of mass when displacement =2cm. c) Find K.E and P.E of the system when displacement =2cm.	5	2
c	I. A three level Laser produce a light of wavelength 550 nm. At what temperature the ratio of the population is half. II. For a ruby laser, the output beam spot diameters are 1mm and 3mm at a distance of 3m and 6m from laser. Calculate the angle of divergence, and area of divergence at 6m away from laser	5	2

P.T.O

	d	A satellite is pushed into a circular orbit 200km above the earth surface where the acceleration due to gravity is 9.2m/sec^2 . Find a) Satellite speed? b) Time period (where the radius of the earth is 6400km.)	5	2
3		<u>Apply the concept and appropriate mathematical relation to find the following:</u>		
	a	a) Find the peak wavelength of black body radiation emitted by each of the following: i) Human body when skin temperature is 35°C . ii) The tungsten filament of a light bulb which operates at 2000K. iii) The Sun which has a surface temperature of about 5800K. b) A cavity whose walls are held at 1900k has small hole of 1mm in dia at what rate does the energy escape through this hole from the cavity interior?	5	3
	b	Molybdenum have a work function of 4.20 eV a) Find the cut off wavelength and cut off frequency for the photoelectric effect. b) What is the stopping potential if the incident light has a wavelength of 180nm.	5	3
	c	i) X-rays of wavelength $\lambda_0 = 0.200000\text{ nm}$ are scattered from a block of material. The scattered X-rays are observed at an angle of 45° to the incident beam. Calculate the scattered wavelength. ii) X-rays having an energy of 300keV undergo Compton scattering from a target the scattered rays are deflected at 37° relative to the incident rays find the Compton shift and the energy of recoiling electron.	5	3
	d	a) Find the ratio of the De-Broglie's wave length of an electron to that of a proton with the same energy? i.e. find $\frac{\lambda_e}{\lambda_p} = ?$ b) Calculate the De-Broglie wavelength of an electron if K.E is 15Kev.	5	3

CONSTANTS:

$h = 6.626 \times 10^{-34} \text{ J.s}$	$e = 1.602 \times 10^{-19} \text{ C}$	$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$	$K = 1.38 \times 10^{-23} \text{ J/K}$
$1\text{amu} = 1.66 \times 10^{-27} \text{ kg}$	$m = 9.11 \times 10^{-31} \text{ kg}$		

Seat No. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
FIRST YEAR (SOFTWARE/ ECONOMICS FINANCE)
SPRING SEMESTER EXAMINATIONS 2024
Batch - 2023

Time: 3 Hours

Dated: 29-07-2024

Max.Marks: 60

ACADEMIC READING & WRITING - EA-115

Note: Attempt all questions in your answer sheets.

Read the passage below and answer the given questions:

The Future of Artificial Intelligence
Barry B. Hughes

The term Artificial Intelligence, or AI, conjures widely different images and expectations for many different people. Some imagine a world filled by autonomous vehicles zipping around without human input. Others may imagine a world where intelligent robots work alongside humans helping to remove much of the drudgery and daily toil from their lives. Some see rapid advances in healthcare and healthcare technologies, enabling humans to live healthier, fitter, and longer lives. Some may see a world where AI becomes the great equalizer, lowering the cost of production and making a wide range of goods available to broad swathes of the population. And yet for some, AI conjures fear and foreboding, a world characterized by mass dislocation of labor and inequality, generating vast social instability. The great fear is that AI comes to surpass human capability with devastating and unknown consequences.

Despite these widely different predictions of future AI and human interaction, AI technologies today remain remarkably limited and narrow, capable of generating only simple outputs like responding to questions, or identifying specific objects within images, or identifying anomalies from complex patterns of data. The world of autonomous agents with intelligence equaling or even exceeding that of humans is still largely a fantasy. And yet today's narrow AI technologies are advancing rapidly, doubling or even tripling their performance over the past five to ten years. AI has been called the "Fourth Industrial Revolution," (Schwab and Samans, 2016a) a recognition its potential impact across a number of important sectors of human development.

AI will have far-reaching effects on the economy; enhancing productivity while at the same time shifting the value-add away from labor and towards capital-intensive machinery and industries. The direct effects on labor are hotly debated. AI technologies are already replacing labor in manufacturing and in some service sectors today, and pessimists suggest this is a harbinger of a broader trend that will lead to massive hollowing out of jobs brought on by automation of tasks and employment. Optimists counter this by pointing out that technology has historically been a net job creator, leading to the development of entirely new industries and specializations previously unavailable. AI will simply free up human capital to pursue more productive and meaningful pursuits, they say. In other sectors, the impact will be similarly broad. Autonomous vehicles could fundamentally restructure transportation infrastructure,

reduce traffic accidents and associated congestion. AI could help drive renewable energy generation and improve demand-side efficiencies, leading to a massive growth in renewable power. AI could personalize education service delivery and produce tools that allow for life-long learning. AI's potential is both wide and deep and only beginning to be realized.

Q. 01: a. Answer the following questions:

[CLO-1] (Marks 10)

- i. What are some of the different images and expectations that people have about AI according to the passage?
- ii. What are the current capabilities of AI technologies mentioned in the passage? Provide specific examples.
- iii. How is AI expected to affect the economy, particularly in terms of productivity and labor?
- iv. Summarize the debate between pessimists and optimists regarding AI's impact on the labor market.
- v. Identify and explain at least three different sectors that could be fundamentally impacted by AI, according to the passage.

b. Write the contextual meanings of the bold italicized words in the text.

[CLO-1] (10 marks)

1. Autonomous
2. Drudgery
3. Foreboding
4. Harbinger
5. Hollowing
6. Pessimists
7. Productivity
8. Specializations
9. Sustainability
10. Transformative

Q. 02: Read the above passage and write a concise summary that captures the main points and key information. Write a summary of the passage in no more than 100 words. Ensure your summary captures the main points about the Artificial Intelligence.

[CLO-3] (10 Marks)

Q. 03: You are tasked with writing a research paper on the impact of renewable energy on environmental sustainability. Properly cite the following sources using the APA/IEEE referencing style:

[CLO-3] (10 Marks)

1. A book titled "Renewable Energy Sources and Climate Change Mitigation" written by Ottmar Edenhofer, Ramón Pichs-Madruga, and Youba Sokona, published in 2011 by Cambridge University Press.
2. An online journal article titled "Assessment of Wind Energy Potential in Coastal Regions of Southern Australia" by Emily Roberts and James Anderson, published in the Journal of Renewable Energy in 2018, volume 35, issue 2, pages 101-113.
3. A conference paper titled "Solar Photovoltaic Technology: Advancements and Future Prospects" by Michael Johnson, presented at the International Conference on Sustainable Energy Technologies in 2020.
4. A webpage from the United Nations Framework Convention on Climate Change (UNFCCC) website, providing statistics on global greenhouse gas emissions for the year 2022.

Q.04: To what extent do you agree or disagree with the text where it says "Optimists counter this by pointing out that technology has historically been a net job creator, leading to the development of entirely new industries and specializations previously unavailable." Justify your answer with examples from your surroundings. (200-250 words)

[CLO-2] (10 Marks)

Q.05. The bar chart below shows the consumption and production of electricity for the top ten countries in the year 2014. Summarize the information by selecting and reporting the main features and make comparisons where relevant. Write at least 150 words.

[CLO-2] (10 Marks)

